



The normal fetal Cavum Septum Pellucidum in MR imaging – New biometric data

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ABSTRACT

Background and purpose: The cavum septum pellucidum (CSP) is an important landmark in the evaluation of the fetal neural axis. A deviation from the ultrasonic normal values may be associated with unfavorable outcomes, and a normal CSP provides reassurance of normal central forebrain development. Today, there is biometric data regarding the normal values for the width of the CSP in fetal ultrasound, but there is no such data for fetal MRI. The aim of this study was to determine the normal values for the measurements of the fetal CSP on MRI.

Materials and methods: We retrospectively examined 307 MRI scans of fetuses between 25 and 41 weeks gestation. Data was collected from the electronic charts of patients who underwent fetal MR imaging at a single tertiary Medical Center. The width and length of the CSP were measured in the axial plane, and the width and height were measured in the coronal plane.

Results: The width and height of the CSP in fetuses tend to decrease starting from the 27th week of gestation onwards. High levels of intraobserver and interobserver agreements were calculated. The sex of the fetus does not appear to influence the biometry of the CSP.

Conclusion: This study provides MRI reference values for the dimensions of the CSP starting from the 25th week of gestation. Knowing the normal values for MRI could provide valuable information for researchers and in the decision-making process in patient's consultations.

1. Introduction

The cave of septum pellucidum (CSP) is an important landmark in the evaluation of the fetal neural axis. It appears as a rectangular or triangular cavity, that contains cerebrospinal fluid (CSF) that filters from the ventricles through the two leaflets of the septum pellucidum [1–3]. Visualization of a normal CSP in the second and third trimester prenatal ultrasound evaluations of the fetal brain provides reassurance of normal central forebrain development [4].

Anatomically, The CSP is defined by the following boundaries: anteriorly – the genu of the corpus callosum (CC); superiorly – the body of the CC; posteriorly – the anterior limb and pillars of the fornix; inferiorly – the rostrum of the CC and the anterior commissure; and laterally – by the laminae of the septum pellucidum [5]. The

development of the CSP can be seen as a secondary process to the development of its bounding structures, the CC and the fornix. As the developing fetal brain grows to its mature form, the CSP grows due to the separation of the fornix and CC from each other [6,7]. The septa begin to fuse in an anterograde fashion at approximately 24 weeks gestation [1,8]. By the age of 6 months the CSP is completely obliterated in 85 % of infants by the fusion of the septa pellucida to form one septum pellucidum [9].

The evaluation of the CSP is vital because its absence could be the result of primary prosencephalic maldevelopment or a secondary process that caused the destruction of the septal leaflets. This condition is associated with a range of disorders such as an agenesis (or dysgenesis) of the CC (ACC), Aicardi Syndrome, schizencephaly, septo-optic dysplasia (SOD); holoprosencephaly (HPE), porencephaly, severe

Abbreviations: CSP, cave septum pellucidum; CC, corpus callosum; GA, gestational age; ICC, intraclass correlation coefficient.

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chronic hydrocephalus, or an acquired fetal brain injury. An isolated agenesis of the CSP may also occur [3,6].

Postpartum, an enlarged CSP (>10 mm) and a persistence of the CSP beyond infancy have been described as significant markers of cerebral dysfunction. They may be accompanied by mental retardation, developmental delay, and neuropsychiatric disturbances [10,11].

There are sonographic studies concerning the normal development and measurements of the CSP [12–15]. The CSP is better visualized by MRI, which is used to confirm and identify associated anomalies too subtle or difficult to detect by US [4], yet to the best of our knowledge, there are no studies regarding the normal development and measurements of CSP in fetal MRI.

The aim of this study was to determine the normal values for the measurements of the CSP in fetal brain MRI.

2. Methods

2.1. Population

We retrospectively examined 307 fetal MRI scans that were performed at a single tertiary referral center between 2005 and 2017. The study population consists of patients between 25 and 41 weeks gestation who underwent MR imaging under different indications. Inclusion criteria for the study were: singleton pregnancy, brain biometry within the normal range and no intracranial anomalies with the exclusion of mild lateral ventricles asymmetry that may be considered a benign finding [16,17], and mild ventricular enlargement (up to 12 mm) which is considered to be a normal variant with favorable outcomes [18,19]. Cases with isolated extracranial anomalies, maternal cytomegalovirus infection without evidence of fetal infection, and normal fetuses with a maternal history of anomalies in previous gestations were also included,

as is acceptable in similar studies [20–23]. In cases involving multiple MRI scans, only one was included.

2.2. Study tools

Data regarding maternal information, fetal MRI measurements and interpretations was collected from the electronic charts. The scans included in the study were re-analyzed for the CSP measurements: width in the axial and coronal planes, length in the axial plane, and height in the coronal plane.

Study measurements were performed manually by a single reader. Four measurements were recorded in each fetus using the following landmarks: Width was defined as the distance between the innermost part of the cavum leaflets where they were at their widest in the axial and coronal planes. Length was defined as the distance between the lower part of the CC and the roof of the third chamber (the fornix) in the axial plane. Height was defined as the distance between the lower border of the anterior part of the CC and the posterior fornix where they meet the thalami in the coronal plane. Examples of measurements are shown in Fig. 1. Measurements were recorded at the 2D slice where the CSP was at its widest, and where the alignment of the axis was correct.

Twenty-one random fetuses were remeasured for the purpose of evaluating intraobserver and interobserver agreements.

2.3. MRI technique

MRI scans performed in the study use a 1.5 T system (Optima scanner, GE Healthcare Technologies, Milwaukee, Wisconsin). Single-shot fast-spin echo T2-weighted sequences in three orthogonal planes were performed using half Fourier technique (NEX = 0.53), with the following parameters: section thickness of 3 mm, no gap, flexible coil (8-

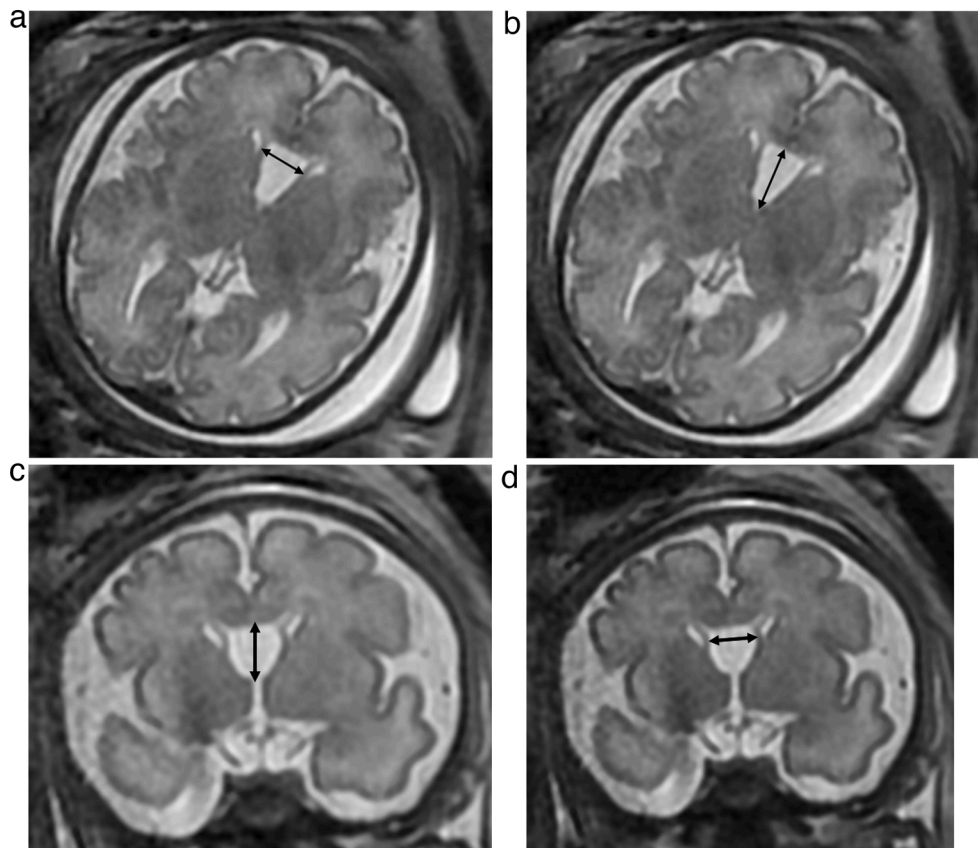


Fig. 1. Fetal brain MRI of the measurements of the CSP in different planes: a. width in the axial plane, b. length in the axial plane, c. height in the coronal plane, d. width in the coronal plane.

channel cardiac coil), field of view (FOV) was determined by the size of the fetal head with a range of 240*240 mm to 300*300 mm, acquisition time between 40 and 45 s, matrix 320/224, TE 90 ms, TR 1298 ms, pixel bandwidth of 122 Hz/pixel; SAR values were between 1.1 and 1.7 W/kg [24].

2.4. Statistical analysis

Continuous variables were evaluated for normal distribution using a histogram and QQ plot and reported as mean and standard deviation (SD).

Categorical variables were expressed as frequencies and percentages. Intraclass correlation coefficient (ICC) [25] was used to evaluate intra-observer agreement.

Pearson correlation was used to study the association between GA and the width, height and length of the CSP.

Analysis of variance (ANOVA) was used to compare difference in measurements between GA groups. All statistical tests were two sided and p -value > 0.05 was considered statistically significant.

Mann Whitney test was used to compare measurements between genders.

Statistical analyses was conducted using the IBM SPSS software (IBM Corp. Released 2016. IBM SPSS Statistics for Windows, Version 24.0. Armonk, NY: IBM Corp.).

2.5. Ethics approval

The study has been approved by the Center's research ethics board.

3. Results

Three hundred and seven patients were measured; 111 fetuses were of male sex, 103 were female, and for 93 fetuses, gender was not recorded. The patient's mean GA was 33.1 weeks (SD 3.38) and the mean maternal age was 31.53 years (SD 5.14).

The cases were divided into eight groups according to their GA (in weeks): 25–26, 27–28, 29–30, 31–32, 33–34, 35–36, 37–38, and 39 weeks and above. The mean values and the reference ranges [approximated value of the 95 % limits (≈ 1.96 SD)] for each GA group in each measurement are presented in Table 1 and in Fig. 2.

Pearson correlation was calculated for the four measurements of the CSP in accordance with the GA. All the results were statistically significant with a correlation of -0.303 ($p < 0.001$) for the axial width, -0.117 ($p = 0.04$) for the axial length, -0.345 ($p < 0.001$) for the coronal width and -0.311 ($p < 0.001$) for the coronal height.

Twenty-one random cases were remeasured by the original reader for the purpose of evaluating intraobserver agreement and measured again by a second reader to evaluate interobserver agreement. ICC was calculated for each measurement, with high levels of agreement, and is presented in Table 2. Strength of agreement was in accordance with Landis and Koch's table of "agreement measures for categorical data"

[25].

An analysis of variance was performed to determine if there is a significant difference between the GA groups for each independent variable (axial width and length and coronal width and height). They were all found to be statistically significant with $p < 0.001$.

A multiple comparison test was performed to compare the width, length, and height of the CSP between male and female fetuses in all GA groups (103 females and 111 males). Out of the 38 comparisons, only two were found to be of statistical significance: coronal height in the 29–30 GA group with $p = 0.049$, and axial length in the 35–36 GA group with $p = 0.015$.

4. Discussion

In this study we attempted to assess the normal values for the measurements of the CSP in fetal brain MRI. So far, to the best of our knowledge, only sonographic data is available.

MRI allows better visualization and analysis of the CSP. therefore, if any abnormality is depicted by prenatal US, an additional MRI can be performed to confirm the diagnosis and identify potential associated anomalies that may be too subtle to detect by US [4,26]. Li et al. [27] examined 23 fetuses with septum pellucidum abnormalities, and found that fetal MRI detected these abnormalities in all 23 patients, while prenatal sonography detected them in only 16 of the patients. In 12 of the fetuses (52 %), additional central nervous system (CNS) findings were seen on MRI and not detected in US. Better interobserver agreement was found among MRI readers in comparison to ultrasound readers.

The CSP is observed beyond approximately 18–20 weeks of gestation under favorable conditions [28], yet the benefits of fetal brain MRI are most prominent in the late second and third trimesters [29]. For that reason and possibly due to the legality of late term abortions for medical reasons in our state, in our center, MRI scans for most indications were performed during the third trimester [30]. Even though the CSP can be demonstrated in MRI as early as 18 weeks GA, we started our study at 25 weeks GA because it was the earliest we were able to find a cohort of fetuses.

Several sonographic studies measured the normal width of the CSP. Serhatlioglu et al. [13] studied 130 fetuses in the second and third trimesters and noted an increase in the size of the CSP from the former to the latter group. Jou et al. [12] studied 608 fetuses between 19 and 42 weeks of gestational age and found that the width of the CSP increases gradually between 19 and 27 weeks gestation, plateaus between 28 and 40 weeks gestation, and ends with a slight decrease towards the 42nd week. Two other sonographic studies have reproduced similar results [14,15]. Until now, there were no studies done, that we know of, regarding the normal measurements of CSP in fetal MRI.

In contrast to a previous studies, our data demonstrated a decrease in the width and height of the CSP of fetuses between 27 and 42 weeks gestation. The trend was statistically significant with a correlation of (-) 0.303 and (-)0.345 respectively.

Table 1

Normal values of the axial width and length and coronal width and height of CSP in different gestational ages; mean and 95 % limits reference range.

GA	N	Axial Width			Axial Length			Coronal Height			Coronal Width		
		mean-1.96SD	mean	mean+1.96SD	mean-1.96SD	mean	mean+1.96SD	mean-1.96SD	mean	mean+1.96SD	mean-1.96SD	mean	mean+1.96SD
25–26	11	1.155	4.181	7.207	1.244	8.205	15.165	1.146	6.744	12.341	0.716	4.062	7.408
27–28	19	3.191	5.188	7.186	5.809	9.701	13.592	5.253	9.233	13.213	3.099	5.236	7.373
29–30	50	1.56	4.41	7.261	5.707	10.205	14.704	4.847	8.126	11.405	1.399	4.439	7.479
31–32	71	1.295	4.535	7.774	6.149	10.221	14.293	3.869	7.844	11.82	1.341	4.54	7.738
33–34	49	0.664	4.238	7.812	5.852	10.181	14.509	4.227	7.727	11.226	0.914	4.444	7.974
35–36	50	1.011	4.008	7.005	4.983	10.113	15.242	3.667	7.154	10.64	0.896	4.046	7.195
37–38	45	0	3.061	6.799	2.168	8.371	14.574	1.283	5.774	10.265	0	2.932	6.502
39+	12	0.106	2.569	5.033	1.636	8.477	15.317	2.956	5.93	8.904	0	2.456	4.992
Total	307												

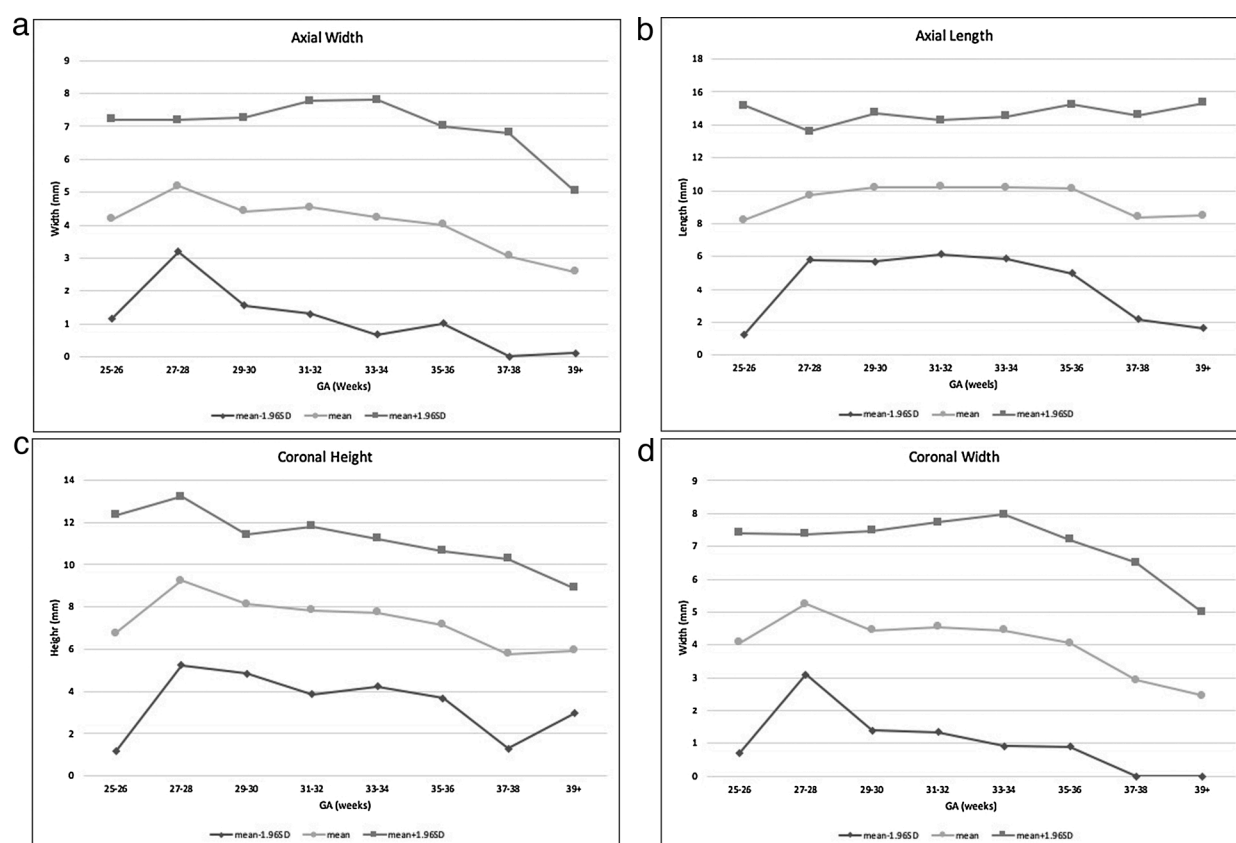


Fig. 2. CSP measurements in different planes in correspondence to the GA group. a. Axial width of the CSP, b. Axial length of the CSP, c. Coronal height of the CSP, d. Coronal width.

Table 2

Calculated interclass correlation coefficient (ICC) for the intraobserver and interobserver agreements.

Measurement	Intraobserver agreement		Interobserver agreement	
	ICC	Strength	ICC	Strength
Axial width	0.930	Almost perfect	0.972	Almost perfect
Axial length	0.732	Substantial	0.769	Substantial
Coronal height	0.928	Almost perfect	0.828	Almost perfect
Coronal width	0.977	Almost perfect	0.985	Almost perfect

Other discrepancies include a smaller mean and larger SD for all GA groups in comparison with previous sonar based studies (SD was larger in comparison to similar size cohorts). These differences could be attributed to the use of a different modality, which further demonstrates the need for this study. In addition, in our study we also measured the CSP's length and height, which were not previously recorded.

A retrospective study that examined the neurodevelopmental outcomes of fetuses detected with narrow and wide CSP in MRI concluded that without an associated fetal abnormality such as aneuploidy or CNS abnormalities, outcomes for both narrow and wide CSP were normal [31]. The diagnosis of wide/narrow CSP was done using the Jou et al. [12] table of sonographic CSP width at various GA. When examining the sonographic data, we found it to be inaccurate for use in MR imaging. Twenty-one of 23 fetuses who were identified as having a narrow CSP in Jou et al.'s study, were found to be well within the 95 % limits of their GA group's values in our study, so they were no longer considered to be with a narrow CSP; which could explain their favorable outcomes. None of the 7 fetuses with wide CSP were within the 95 % limits for their GA group in our study, however, their outcomes were favorable in the study.

Out of 38 comparisons made between male and female fetuses, only

two outcomes have met the significance threshold; this could be attributed to the multiple comparisons themselves. Taking this attribution into account, and with and the multiple non-significant results we can conclude that there is no difference between male and female fetuses regarding the normal development of the CSP.

The limitations of our study are mostly due to the study's population and retrospective design. We intended to reach conclusions regarding the normal and healthy population of fetuses in an MRI study, yet it is unethical or at least problematic to subject a healthy woman and fetus to an MRI scan (exposure to noise, sedation) solely for the sake of research. Therefore, for the purpose of our study, we took near normal fetuses and deduced from them the normal values for the healthy population. This deduction is a form of selective sampling bias. To minimize the effect of the bias we selected only fetuses that were without intracranial malformations, as they were defined in the study population section; this method is the common practice in fetal MRI studies [20–23].

Further limitations are the number of cases in the early and late GA groups as we were limited in the number of fetuses available for measuring, leading to a relatively large SD. We chose to include them in the study because we wanted to apply our data as widely as possible. Finally, in a third of cases the sex of the fetus was not recorded which decreased the statistical power of the comparison between male and female fetuses.

The strengths of our study are its large sample size, with a cohort of over 300 fetuses and with over 1200 separate measurements; the wide range of gestational ages (25–41 weeks); the measurement of the width of the CSP in both the axial and coronal planes; the inclusion of additional measurements (height and length); the high level of intraobserver and interobserver agreements; and being the first study to measure the CSP on MRI.

5. Conclusions

Nowadays, there are studies that describe the normal values for the width of the CSP in fetal ultrasound, but no such studies exist for fetal MRI. A deviation from the ultrasonic normal values can be associated with unfavorable outcomes, and an MRI is performed to confirm or refute a sonographic or clinical suspicion, being a more accurate modality.

This study provides MRI reference values for the dimensions of the CSP starting from the 25th week of gestation. Knowing the normal values for MRI could provide valuable information for researchers and in the decision-making process in patient's consultations.

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CRediT authorship contribution statement

Itay Kertes: Methodology, Investigation, Writing - original draft, Visualization. **Dana Hoffman:** Validation. **Orr Yahal:** Resources. **Michal Berkstadt:** Resources. **Omer Bar-Yosef:** Resources. **Osnat Ezra:** Writing - review & editing. **Eldad Katorza:** Conceptualization, Methodology, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors report no declarations of interest.

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